

Effects of sampling microhabitats with low coverage within the STAR/AQEM macroinvertebrate sampling protocol on stream assessment

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Abstract

In the STAR/AQEM protocol microhabitats covering less than 5% of the sampling area were neglected. Driven by an ongoing discussion on the importance of these underrepresented microhabitats we tested the influence of sampling them. We investigated 48 streams representing 14 different stream types from all over Germany. Macroinvertebrates of underrepresented microhabitats were sampled in addition to the STAR/AQEM protocol. To ensure the method remains feasible in routine monitoring programmes the total sampling and sorting effort of additional sampling was limited to 20 min. Particularly those taxa were picked, which were not recognised during the routine STAR/AQEM sorting.

To identify the effect of additional sampling on stream assessment results, we calculated the stream type-specific Multimetric Index (MMI) with the “main” and the “main + additional” data for each sample. The mean and median difference in MMI values between “main” and “main + additional” samples was 0.02 and 0.01, respectively. In seven of 48 samples (14.6%) a different ecological quality class was calculated with the “main + additional” dataset. Regarding common metrics within the MMI as well as intercalibration metrics differences between “main” and “main + additional” samples were analysed. The values differed most in richness metrics (e.g., number of EPTCBO Taxa, number of Trichoptera Taxa). The results of the present study show that additional sampling of underrepresented microhabitats could alter multimetric assessment results.

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Introduction

The STAR/AQEM protocol (Furse et al., 2006; STAR consortium, 2003) is the result of two successive EU-funded projects, which aimed to develop a standar-

dised macroinvertebrate sampling protocol and assessment system for running waters (Furse et al., 2006; Hering et al., 2003). STAR/AQEM sampling is based on 20 sampling units, which are distributed according to the estimated microhabitat distribution at a sampling site. Microhabitat coverage is estimated in 5% intervals and only those microhabitats which cover at least 5% at the sampling site (50 m for streams, 100 m for rivers) are sampled. Each sampling unit (25 cm × 25 cm) is sampled using a kick sampling method with a hand net

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(25 cm × 25 cm frame; 500 µm mesh size). Microhabitats covering less than 5% are neglected based on the hypothesis that these microhabitats will have no effect on river assessment. In the present study, those microhabitats are defined as “underrepresented” microhabitats.

There is much discussion about neglecting these underrepresented microhabitats, with some people arguing that those microhabitats are inhabited by additional taxa or taxa that are particularly important in stream assessment. In Germany, this discussion is strongly supported by water managers who demand incorporating underrepresented microhabitats in the German standard monitoring programme. Their underlying hypothesis being that additional sampling of underrepresented microhabitats will lead to additional taxa and therefore “better” assessment results.

In our study, we test the influence of these additional taxa on selected metrics and assessment results by sampling underrepresented microhabitats in addition to the slightly modified STAR/AQEM protocol according to Haase, Lohse, et al. (2004). Because our sampling includes different ecoregions, inhabited by a different number of taxa (Illies, 1978), the above hypothesis is also tested with regard to ecoregions. However, the effort for this additional sampling must be restricted to limit costs of the German monitoring programme. The sampling design of this study had to take this restriction into account and was therefore mainly a practical approach.

The main questions addressed in this study are: (a) Does additional sampling effect the number of taxa and/or taxa composition? (b) Does additional sampling effect metric and/or assessment results? and (c) Does the number of additional taxa and/or changes in assessment results differ between ecoregions?

Material and methods

Main sampling

Macroinvertebrate samples were taken at 48 sampling sites. The sites were distributed all over Germany and included 14 of 24 German stream types from two ecoregions according to Illies (1978): Central Highlands (No. 9) and Central Plains (No. 14). Sampling took place in the summer of 2004 (the recommended sampling season for medium to large size rivers) following the slightly modified STAR/AQEM protocol according to Haase, Lohse, et al. (2004), which is the common standard protocol in Germany. This protocol basically corresponds to the original STAR/AQEM protocol (Furse et al., 2006; STAR consortium, 2003) with one modification: the sample material was separated over a 2 mm sieve and only the >2 mm fraction

remaining in the sieve was considered for further treatment. All samples were sorted in the lab. In the field, the sample material was scanned for selected taxa, which may be damaged by further treatment of the sample material. These taxa were picked and separated for determination. A welcome side effect of this procedure was that the person taking the sample got a rough overview of the taxa in the sample material. All determinations were done by the same team according to the taxonomic level (mostly species level) defined by Haase, Sundermann, and Schindehütte (2006a) to ensure comparable determination results. For the purpose of this study, we defined the resulting taxa list as “main” sample.

Additional sampling

After the “main” sample, an additional sample was collected, where all microhabitats covering less than 5% were sampled by kick sampling or manual searching (e.g., by picking organisms from woody debris). All of these samples were pooled for further treatment. In five cases no underrepresented microhabitats covering less than 5% were recorded. At these sites additional sampling was performed in microhabitats with the lowest estimated coverage. Due to the requirements of the German water managers (minimal time effort and costs) the total effort for this additional sampling was restricted to 20 min (sampling + sorting). Only aquatic stages of macroinvertebrates were sampled. The organisms from the additional sample were sorted and removed qualitatively in the field (live sorting). We particularly collected taxa that were not recognised during the routine scanning of the sample material in the field (see above) to increase the number of additional taxa. After determining the additional specimens in the lab the resulting taxa list of the additional sample (from hereon defined as “additional” sample) was added to the taxa list of the “main” sample. For the purpose of this study, we defined the resulting taxa list as “main + additional” sample.

Data analyses

For each “main” and “main + additional” sample the German stream type-specific Multimetric Index (MMI), the four most common core metrics within the MMI (Table 1), namely: (1) the German Fauna Index (“GFI”, Lorenz, Hering, Feld, & Rolaufts, 2004; Meier et al., 2006), (2) the percentage of Ephemeroptera, Plecoptera and Trichoptera taxa (“EPT Taxa [%]”), (3) the number of Ephemeroptera, Plecoptera, Trichoptera, Coleoptera, Bivalvia and Odonata (“No. EPTCBO Taxa”), and (4) the number of Trichoptera taxa (“No. Trichoptera Taxa”), and the ecological quality class (EQC) were

Table 1. Core metric composition of the stream type-specific MMI

Core metrics	Stream types													
	2.1	2.2	3.2	4	9	9.1	10	12	15	17	20	21_N	21_S	23
GFI	X	X	X	X	X	X	–	X	X	X	–	X	X	–
EPT Taxa (%)	X	X	X	X	X	X	–	X	X	X	–	X	X	X
No. EPTCBO Taxa	X	X	X	X	X	X	–	–	–	–	–	–	–	–
No. Trichoptera	–	–	–	–	–	–	–	X	X	X	–	–	–	–
Potamontypic Index	–	–	–	–	–	–	X	–	–	–	X	–	–	–
Rheoindex	X	X	X	X	–	–	–	–	–	–	–	–	–	–
Oligosaprobic Taxa (%)	–	–	–	–	–	–	–	–	–	–	–	–	–	X
Littoral Taxa (%)	–	–	–	–	–	–	–	–	X	X	–	–	–	–
Phytal Taxa (%)	–	–	–	–	–	–	–	–	–	–	–	X	X	–
Pelal Taxa (%)	–	–	–	–	–	–	–	–	X	–	–	–	–	X
Metarhithral Taxa (%)	–	–	–	–	X	–	–	–	–	–	–	–	–	–
Epipotamal Taxa (%)	–	–	–	–	–	–	–	–	–	–	–	–	–	X
Metapotamal Taxa (%)	–	–	–	–	–	–	–	–	–	–	–	–	–	X
<i>N</i>	1	2	3	2	3	14	2	6	3	3	2	1	2	4

For further information see [Meier et al. \(2006\)](#). *N* = number of streams sampled. Stream types according to [Pottgiesser and Sommerhäuser \(2004\)](#). Explanation of stream types: 2.1, small streams in the alpine foothills; 2.2, mid-sized streams in the alpine foothills; 3.2, mid-sized streams in the Pleistocene sediments of the alpine foothills; 4, large streams in the alpine foothills; 9, mid-sized siliceous cobble/boulder bottom streams in lower-mountainous areas; 9.1, mid-sized streams in calcareous lower-mountainous area; 10, very large gravel-dominated rivers. 12, mid-sized streams with organic substrates; 15, mid-sized to large sand-bottom streams in the lowlands; 17, mid-sized to large gravel-bottom streams in the lowlands; 20, very large sand-dominated rivers, 21_N, lake outflows in the lowlands; 21_S, lake outflows in the alps and alpine foothills; 23, backwater and brackish water influenced Baltic Sea tributaries.

calculated by using the ASTERICS software ([AQEM/STAR Ecological River Classification System, Version 3.01, AQEM consortium, 2006](#)).

The German MMI comprises a stream type-specific set of core metrics ([Table 1](#)). As these core metrics are scored either as continuous (e.g., “No. EPTCBO Taxa”) or percentage scores (“EPT Taxa [%]”), their values are normalised to a range 1 (representing the reference condition) to 0 (representing bad status) ([Böhmer et al., 2004](#)). This normalisation simplifies the combination of metric results and calculation of the final MMI assessment result. Since class sizes are equal for all metrics the following classes are defined and applied for stream assessment in Germany: high (1) ≥ 0.8 , good (2) ≥ 0.6 to < 0.8 , moderate (3) ≥ 0.4 to < 0.6 , poor (4) ≥ 0.2 to < 0.4 , and bad (5) < 0.2 .

Intercalibration metrics ([Buffagni et al., 2006](#)) are important metrics for harmonising European stream assessment methods between countries. Thus, we also calculated European stream assessment intercalibration metrics using the ASTERICS software. These metrics were: (1) ASPT, (2) $\text{Log}_{10} (\text{Sel_EPTD} + 1)$, (3) 1-GOLD, (4) total number of families, (5) number of EPT families, and (6) Shannon–Wiener diversity. In contrast to the core metrics described above, the intercalibration metrics were not normalised because stream type-specific reference values for a high and a poor/bad ecological status were unknown ([Böhmer et al., 2004](#)).

We calculated a Wilcoxon-Test to examine whether core metric results react differently on additional sampling. We used Mann–Whitney’s *U*-test to examine whether there are significant differences concerning number of taxa and MMI values in different ecoregional groups. All statistical analyses were performed with the STATISTICA 6.1 software package ([StatSoft, 2002](#)).

Results

Number of taxa and individuals

The median number of organisms found in the “main” samples is 1848 ([Table 2](#)). The number of taxa ranges from 7 to 54 with a median of 34 ([Fig. 1](#)). The number of new taxa found in the additional sample ranges from 0 to 7 ([Fig. 1](#)). The median number of taxa collected in the additional sample is 8. On average, additional sampling results in two new taxa (6.6%), which are not included in the “main” sample ([Table 2](#)).

Taxa composition

The taxa composition of “main” and “additional” samples is quite similar. [Fig. 2](#) shows the relative abundance of different taxonomic groups for “main” and “additional” samples, respectively. The maximum

Table 2. Number of taxa and individuals including median values from “main” and “additional” sample

Sample no.	Stream type	Ecoregion	Main sample			Additional sample		
			Abundance (ind/m ²)	Number of taxa	Number of individuals	Number of taxa	Number of new taxa	Percentage of new taxa
1	2.1	No. 9	1397	34	26	4	2	5.6
2	2.2	No. 9	3134	42	37	13	7	14.3
3	2.2	No. 9	1847	32	18	7	1	3.0
4	3.2	No. 9	690	47	60	18	4	7.8
5	3.2	No. 9	2225	43	34	14	6	12.2
6	3.2	No. 9	1681	46	22	11	2	4.2
7	4	No. 9	3755	34	27	6	1	2.9
8	4	No. 9	1901	30	65	16	5	14.3
9	9	No. 9	3494	35	27	14	6	14.6
10	9	No. 9	1355	47	25	11	3	6.0
11	9	No. 9	10506	54	20	8	0	0.0
12	9.1	No. 9	2353	50	29	5	3	5.7
13	9.1	No. 9	2700	20	17	2	0	0.0
14	9.1	No. 9	1551	36	12	4	2	5.3
15	9.1	No. 9	1428	41	16	4	2	4.7
16	9.1	No. 9	984	37	14	8	5	11.19
17	9.1	No. 9	3662	31	26	12	4	11.4
18	9.1	No. 9	4066	44	27	13	2	4.3
19	9.1	No. 9	9126	30	6	4	0	0.0
20	9.1	No. 9	2274	53	49	7	1	1.9
21	9.1	No. 9	6248	50	20	15	5	9.1
22	9.1	No. 9	700	33	15	7	4	10.8
23	9.1	No. 9	9751	44	18	8	3	6.4
24	9.1	No. 9	905	28	20	14	7	20.0
25	9.1	No. 9	1187	40	33	16	5	11.1
26	10	No. 9	2474	40	13	8	1	2.4
27	10	No. 9	1564	17	24	7	1	5.6
28	12	No. 14	887	31	28	7	2	6.1
29	12	No. 14	671	43	20	10	1	2.3
30	12	No. 14	3617	36	19	8	3	7.7
31	12	No. 14	572	22	4	4	2	8.3
32	12	No. 14	717	23	5	2	1	4.2
33	12	No. 14	1849	24	3	3	3	11.1
34	15	No. 14	2794	26	9	4	1	3.7
35	15	No. 14	1197	27	28	13	7	20.6
36	15	No. 14	2038	35	22	12	3	7.9
37	17	No. 14	1831	28	43	17	5	15.2
38	17	No. 14	203	31	24	8	3	8.8
39	17	No. 14	1175	44	24	14	1	2.2
40	20	No. 14	4867	7	15	5	1	12.5
41	20	No. 14	1909	24	23	8	1	4.0
42	21_N	No. 14	1928	42	17	6	1	2.3
43	21_S	No. 9	4254	33	24	9	3	8.3
44	21_S	No. 9	1928	34	48	14	7	17.1
45	23	No. 14	4254	24	16	6	2	7.7
46	23	No. 14	230	20	15	7	2	9.1
47	23	No. 14	1766	18	13	2	0	0.0
48	23	No. 14	619	53	14	10	0	0.0
Median			1848	34	21	8	2	6.6

Number of additional taxa refers to taxa exclusively collected in the “additional” sample. Percentage of new taxa refers to the overall number of taxa at a site. Ecoregions according to Illies (1978): No. 9 = Central Highlands; No. 14 = Central Plains. Stream types according to Pottgiesser and Sommerhäuser (2004).

difference is 4% observed in Coleoptera and Odonata. In both types of samples (“main” and “additional”) more than half of the taxa belong to the orders Trichoptera, Coleoptera and Ephemeroptera (Fig. 2).

Effects of additional sampling on metric results and EQCs

Fig. 3 shows the MMI results and EQCs of “main” and “main + additional” samples ordered by number of additional taxa. In seven of 48 cases (14.6%) the different MMI values lead to different EQCs (Fig. 3: ellipses). In all seven cases the differences cause a shift to a neighbouring EQC. In five cases we observed a better EQC, in two cases a worse EQC after additional sampling. These shifts coincide with mainly positive deviations in MMI values.

The median difference in MMI values of the “main” sample and the “main + additional” sample is 0.01

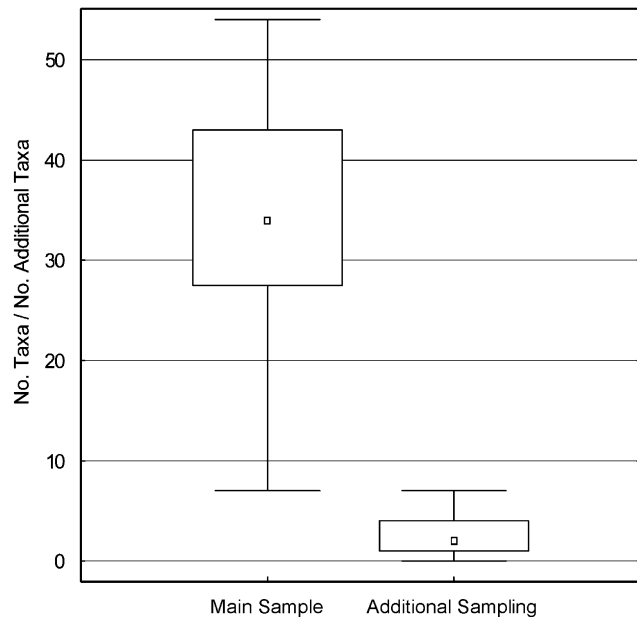
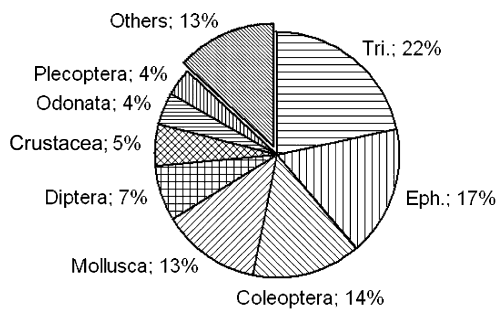


Fig. 1. Box–Whisker plots showing the number of taxa in the “main” sample (left plot) and number of additional taxa found in the “additional sampling” material (right plot). $N = 48$.



(Fig. 4; mean difference is 0.02, data not shown). The median differences in “GFI” and “EPT Taxa [%]” scores are close to zero (Fig. 4). But like the MMI, both metrics show differences in absolute values indicating an impact of additional sampling (median values of absolute differences in MMI: 0.01; “GFI”: 0.02; “EPT Taxa [%]”: 0.01, data not shown). In “No. EPTCBO Taxa” the median difference in score is 0.08, which is quite high considering that a difference of 0.2 will always lead to a different EQC. The differences in “No. EPTCBO Taxa” scores are significantly higher than the differences in MMI, “GFI” and “EPT Taxa [%]” scores (Wilcoxon, $p < 0.01$). In several samples, the differences in “No. Trichoptera Taxa” scores are also high. However, the median difference in these metrics is zero. Because the metric “No. Trichoptera Taxa” is only used for assessment in few stream types, the data set for this analysis was restricted to 12 samples and should therefore be interpreted with caution.

The influence of additional sampling on intercalibration metrics is minimal. We only observed differences in original (not normalised) metric values and these differences are low (Table 3).

Impact of ecoregions on additional sampling

As shown in Fig. 5, there are only minor differences in the two compared ecoregions in median values of MMI, “GFI” and “EPT Taxa [%]”. The slight differences shown in the box plots are not significant (Mann–Whitney’s U -test, $p > 0.05$). The metrics “No. EPTCBO Taxa” and “No. Trichoptera Taxa” are not considered, as they are only core metrics in one of the two ecoregions.

Discussion

Impact of additional sampling on taxa composition

Additional sampling increases the number of individuals and taxa slightly while taxa composition (% of

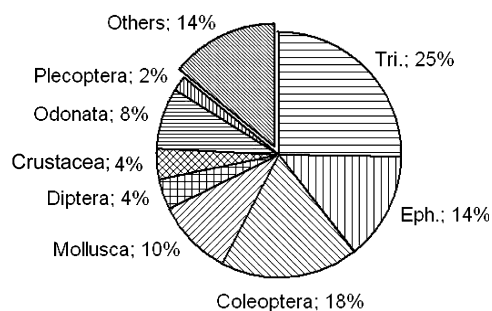


Fig. 2. Taxa composition of the “main” sample (left pie chart) and the “additional” sample (right pie chart). Eph. = Ephemeroptera; Tri. = Trichoptera.

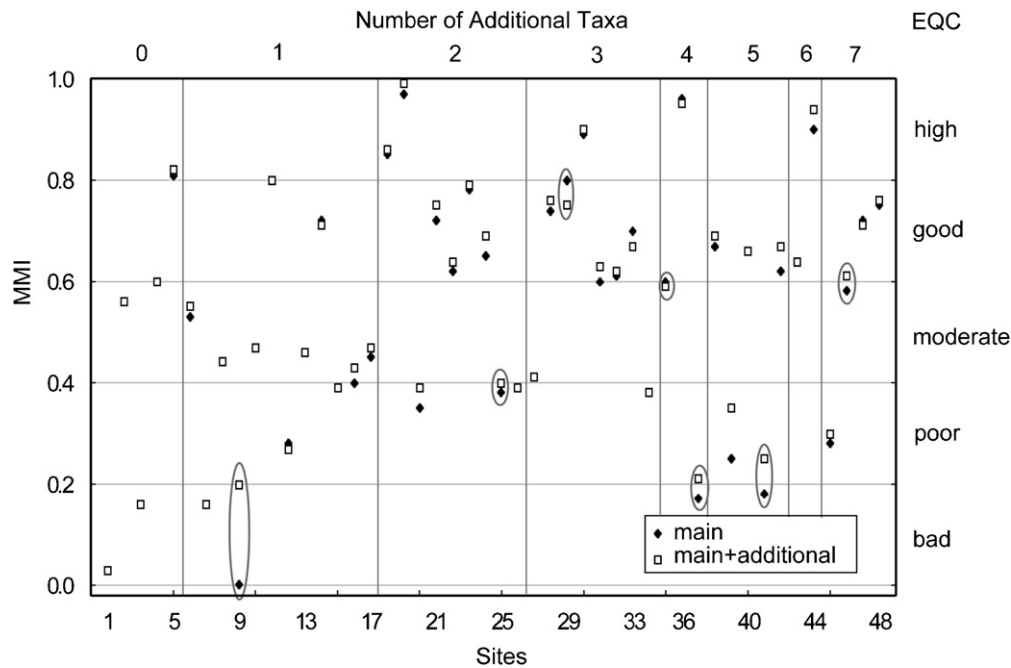


Fig. 3. MMI results of the “main” and the “main + additional” samples. Samples were sorted in ascending order by number of new taxa within the “additional sample”.

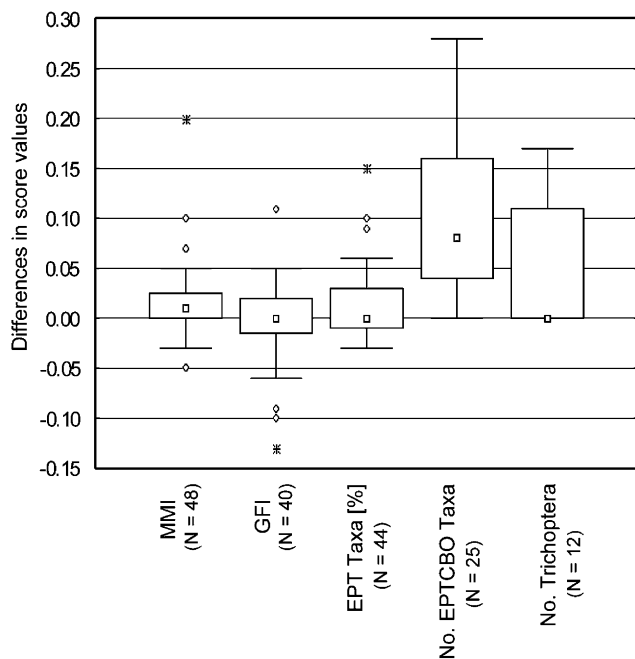


Fig. 4. Box-Whisker plots showing the differences in MMI and core metric results. The differences are calculated as follows: difference in score values = score value (“main + additional” sample) – score value (“main” sample). The stream type-specific MMI can comprise a different set of core metrics. Therefore given “N” values are different for each core metric.

higher taxonomic units) of “main” and “main + additional” sampling is quite similar. These aspects are interesting with regard to the sampling design: Within the additional sampling predominantly those taxa are

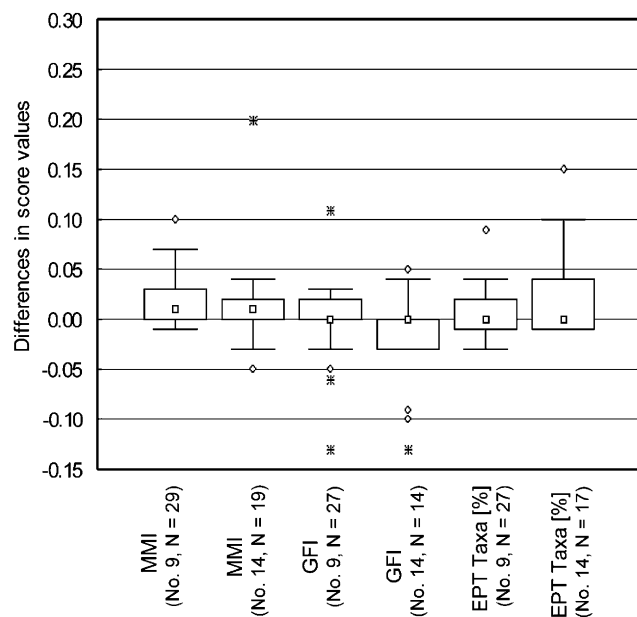
sampled, which are not detected in the “main” sample. Although investigators focus on finding taxa not found in the “main” sample, only a few of the taxa collected are actually new to the sample. The reason for this might be the general problem of identifying taxa in the field (Carter & Resh, 2001; Rawer-Jost, 2001). Therefore, taxonomic composition remains nearly the same as in the “main” sample.

Impact of additional sampling on stream assessment and its effect in different ecoregions

We expect a higher number of taxa to be present in streams of the more southern Central Highlands (No. 9) than the more northern Central Plains (No. 14) (Illies, 1978; Jacobsen, Schultz, & Encalada, 1997; Vannote & Sweeney, 1980). Therefore, we would also expect to find more new taxa in the “additional” samples from the Central Highlands. However, in our study the observed differences in additional taxa are not significant, probably because in both ecoregions the absolute number of additional taxa was comparably low. Interestingly, Verdonschot (2006) questions the basic principle that the number of taxa rises with decreasing latitude. His analysis of a dataset of 876 “reference samples”, taken in 13 different European countries, revealed no relation between average number of taxa and latitude (Verdonschot, 2006). If there is really no relationship between number of taxa and latitude, at least for routine benthic invertebrate samples which are taken for stream assessment purposes, our expectation

Table 3. Differences in intercalibration metrics (ICMs) between “main” and “main + additional” samples

	ASPT	Log10(Sel_EPTD + 1)	1-Gold	No. of families	No. of EPT families	SW diversity
<i>Differences in relative values</i>						
Min	−0.349	0.000	−0.019	0	0	−0.006
Max	0.500	0.440	0.031	6	1	0.147
Mean	0.036	0.024	0.003	1.42	0.04	0.028
SD	0.159	0.067	0.007	1.40	0.20	0.026
Median	0.000	0.000	0.002	1	0	0.024
<i>Differences in absolute values</i>						
Mean	0.107	0.024	0.005	1.42	0.04	0.028
SD	0.121	0.067	0.006	1.40	0.20	0.026
Median	0.067	0.000	0.002	1	0	0.024

**Fig. 5.** Box-Whisker plots showing the differences in MMI and core metric results in dependence of the two investigated ecoregions (according to Illies (1978): No. 9 = Central Highlands; No. 14 = Central Plains).

to find more new taxa within the “additional” samples from the Central Highlands than from the Central Plains is incorrect. However, the analyses of Verdonschot (2006) are based on different sampling methods, and although taxonomically adjusted, still on different taxonomic levels. In several studies both the influence of sampling method and taxonomic resolution on the number of taxa are clearly demonstrated (Clarke, Davy-Bowker, et al., 2006; Friberg et al., 2006; Vlek, Sporka, & Krno, 2006). Therefore, it still remains unclear whether there is a relation between ecoregions and number of taxa. In contrast, an unpublished analysis of Brändle et al. (in preparation) correlating the taxa and ecoregions from Illies (1978) clearly shows a relation between latitude and number of lotic taxa (decreasing number of taxa from south to north). This

corresponds to the widely accepted hypothesis that the diversity of benthic invertebrates is on average higher in southern Europe than in northern Europe. Regarding the impact of additional taxa towards stream assessment results, the slightly, if not significantly higher (*U*-test, $p = 0.22$) number of additional taxa found in “additional” samples from the Central Highlands obviously result in the same relative impact of additional sampling, as is the case in the Central Plains. Thus, no significant difference in metric and assessment results was observed between the two compared ecoregions (Fig. 5).

However, the ecoregions level impact of additional sampling varies between metrics: while the intercalibration metrics remain more or less unaffected, some of the core metrics used for stream assessment in Germany show different results. This is mainly true for richness metrics, e.g., “No. EPTCBO Taxa” and “No. EPT Taxa”. Among the intercalibration metrics the only richness metric is “No. EPT Families”, which measures richness at a high taxonomic level. On the other hand, the “GFI” and “EPT Taxa [%]” show little reaction to additional sampling. The tendency that metrics, which count taxa react more sensitively is also demonstrated by Clarke, Lorenz, et al. (2006).

The resulting differences in mean MMI values of 0.02 seem to be low. However, it is very difficult to estimate the true meaning of these differences: in the German assessment system MMI values range from 0 to 1 and the class boundary is set at 0.2 MMI point intervals. Therefore a change in MMI values of 0.2 will always lead to a different EQC. Because there is a mathematically complicated non-linear relationship between the calculated differences in MMI values and the 0.2 EQC interval, estimating the impact of a certain change in MMI value is difficult. For example, a hypothetical difference in MMI values of 0.02 is not equivalent to 10%. Nonetheless, in our study a mean difference of 0.02 in MMI values leads to a different EQC at seven of 48 sites (14.6%). This indicates that there is a recognisable impact of additional sampling on stream assessment results.

Impact of modifications and variability of the STAR/AQEM method on assessment results

The German assessment system was mainly developed and calibrated based on original STAR/AQEM samples. A recent change to the German version of the STAR/AQEM protocol, calls for operators to include an “additional” sample of underrepresented microhabitats. As shown in our study, sampling of underrepresented microhabitats as “additional” samples, can lead to higher or lower MMI values. Therefore, it is not possible to create a correction factor for the assessment system. It is therefore still unknown whether this alteration in methodology will lead to better, or worse assessment results, or whether assessments will remain largely unaffected.

These findings confirm several recent studies on the influence of modifications or variability on the STAR/AQEM methodology and assessment approaches: there are, e.g., significant deviations in metric results caused by different investigators, or observed in replicate sampling or sampling in different seasons (Clarke & Hering, 2006; Clarke, Davy-Bowker, et al., 2006; Clarke, Lorenz, et al., 2006; Friberg et al., 2006; Haase, Pauls, Sundermann, & Zenker, 2004; Lorenz & Clarke, 2006; Sporka, Vlek, Bulánková, & Krno, 2006). The same is true concerning precision and errors (Haase, Murray-Bligh, et al., 2006). On the other hand, all these studies clearly indicate that observed differences are not a weakness of the STAR/AQEM protocol but a general problem in sampling invertebrates in streams. The main reason for this is the non-random distribution of invertebrates in streams: they are usually distributed in heterogeneous patches (Barmuta, 1989; Downes, Lake, & Schreiber, 1993; Pringle et al., 1988; Townsend, 1989). Therefore, although a highly standardised protocol like the STAR/AQEM is used for several samples at a single site, the species composition of a certain sampling site may differ significantly due to their natural variability.

The main goal of a multimetric assessment approach is thus to lower the impact of this variation on assessment results. The greatest difficulty in achieving this is that different metrics react differently to patchiness or modifications in methodology. So, while the STAR/AQEM approach is among to the most standardised protocols in river assessment in Europe, its precision is still not optimal due to natural variability and its varying effect on different metrics.

Like other methods the STAR/AQEM method will evolve over time and remains subject to future modifications and adaptations like the additional sampling procedure presented in our study. As shown in our investigation and some of the studies mentioned above, the STAR/AQEM protocol generally reacts quite sensitive to changes in methodology. However, considering that a certain degree of sensitivity is essential in

a stream assessment scheme, this sensitivity to changes in methodology should not per se be regarded as a negative feature. Without sufficient sensitivity it is impossible to detect changes towards different EQCs. Water managers should take into account, that any kind of changes to highly standardised protocols like the STAR/AQEM method or RIVPACS (Wright, Armitage, & Furse, 1989) may lead to varying assessment results and finally to a different ecological classification of running waters.

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